

Doping Stability Report -- Ca in KCoO2 at 300 C

Generated: 2026-05-30 14:28 Host: KCoO2 Dopant: Ca

Executive Summary

Ca preferentially occupies the K layer. The formation energy difference between the two layers is +0.356 eV, strongly favouring the K site. MD confirms this preference: on the K site the dopant MSD plateau slope is -0.00089 Å²/ps (stable), compared to -0.00209 Å²/ps on the Co site.

Site	E _f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
K	-3.294	+1	No	PASS	FAIL	STRUCTURAL COLLAPSE
Co	-2.937	-1	No	PASS	FAIL	STRUCTURAL COLLAPSE

Ca at the K layer -- STRUCTURAL COLLAPSE

Charge situation

Ca replaces K with a charge surplus of +1. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E _f	-3.2935 eV	PASS	< 1.0 eV to pass
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E_f = -3.294 eV (approximate -- charge surplus not modelled in CHGNet). This value is exceptionally large and negative, indicating that Ca incorporation is strongly thermodynamically driven at the K site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	-0.00089 Å²/ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.528 Å²	----	
Max displacement	0.807 Å	----	
Coordination (min/mean/max)	1 / 3.8 / 6	FAIL	relaxed = 5, tolerance +/-1
Mean bond length (MD vs relaxed)	2.325 Å (relaxed: 2.3)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = -0.00089 Å²/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.528 Å²; maximum displacement from starting position = 0.81 Å.

Coordination fluctuated between 1 and 6 (mean 3.8), deviating by more than +/-1 from the relaxed value (5). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.325 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STRUCTURAL COLLAPSE

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact

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rather than a true structural collapse. The Ca atom barely moves (MSD = 0.528 Å²) but the coordination count fluctuates because the bond length at the K site is close to the analysis cutoff. Consider increasing coordination_cutoff_A and re-running analysis.

? *WARNING: Charge surplus (+1) cannot be modelled in CHGNet. Running single substitution without compensation. E_f is approximate.*

Ca at the Co layer -- STRUCTURAL COLLAPSE

Charge situation

Ca replaces Co with a charge mismatch of -1. To restore charge neutrality, 1 Na atom(s) were removed from the supercell (furthest from the dopant site). The formation energy accounts for the Na-vacancy term.

Formation Energy (thermodynamic stability)

Formation energy E _f	-2.9374 eV	PASS	< 1.0 eV to pass
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E_f = -2.937 eV. This value is strongly negative, indicating that Ca incorporation is thermodynamically very favorable at the Co site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	-0.00209 Å ² /ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.105 Å ²	----	
Max displacement	0.891 Å	----	
Coordination (min/mean/max)	1 / 4.6 / 5	FAIL	relaxed = 5, tolerance +/-1
Mean bond length (MD vs relaxed)	2.263 Å (relaxed: 2.2)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = -0.00209 Å²/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.105 Å²; maximum displacement from starting position = 0.89 Å.

Coordination fluctuated between 1 and 5 (mean 4.6), deviating by more than +/-1 from the relaxed value (5). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.263 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STRUCTURAL COLLAPSE

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Ca atom barely moves (MSD = 0.105 Å²) but the coordination count fluctuates because the bond length at the Co site is close to the analysis cutoff. Consider increasing coordination_cutoff_A and re-running analysis.

? *WARNING: Charge deficit (-1) cannot be compensated in CHGNet (would require adding K interstitials or oxidising Co). Running single substitution without compensation. E_f is approximate.*

Site Preference Comparison

The K site is preferred by 0.356 eV over the Co site. Formation energy: -3.294 eV (K) vs -2.937 eV (Co).

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Caveats and Limitations

? CHGNet is charge-neutral.

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch $\neq 0$) are approximate even when structural compensation defects are included.

? No Freysoldt/Kumagai correction.

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? Metal-rich reference state.

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? MD timescales are short (25 ps default).

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? Energy rankings are more trustworthy than absolute values.

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.